ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

DISTANCE LEARNING PROGRAMME
(Academic Session : 2024 - 2025)

JEE (Advanced)
TEST # 11
24-11-2024

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Review (Unit Test # 06, 07 & 08)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	C	A	A		
SECTION-I (ii)	Q.	5	6	7			
	A.	C,D	A,C	A,B,C			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	C	A	C	D		
SECTION-II	Q.	1	2	3	4	5	6
	A.	4	1	4	5	3	2

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	C	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C	B,C,D	A,C,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	A	C	C	A		
SECTION-II	Q.	1	2	3	4	5	6
	A.	5	9	8	1	4	1

PART-3 : MATHEMATICS

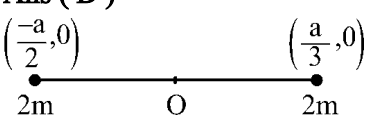
SECTION-I (i)	Q.	1	2	3	4		
	A.	D	B	C	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C,D	A,B,C,D	B,D			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	A	B	C	A		
SECTION-II	Q.	1	2	3	4	5	6
	A.	1	8	8	7	2	8

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

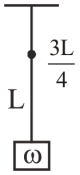
1. Ans (D)



Consider total mass of square = 4m

$$x_{cm} = \frac{-2m\left(\frac{a}{2}\right) + 2m\left(\frac{a}{3}\right)}{4m} = -\frac{a}{12}$$

2. Ans (C)



$$F = \left[\frac{3\omega}{4} + \omega_1\right]$$

$$\text{Stress} = \frac{F}{S} = \frac{\left[\omega 1 + \frac{3\omega}{4}\right]}{S}$$

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3. **Ans (A)**

By analogy with SHM, $\left(\frac{dI}{dt}\right)_{max}$ is analogous to a_{max}.

Since $a_{max} = \omega^2 A$

$\therefore \left(\frac{dI}{dt}\right)_{max} = \omega^2 q_0 = \frac{q_0}{LC}$

4. **Ans (A)**

$k[100 - T_1] = 2k[T_1 - T_2] = k[T_2 - 20]$

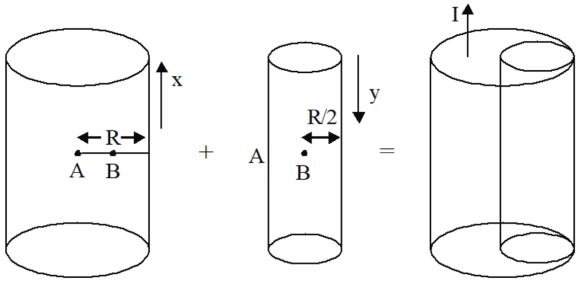
$100 - T_1 = 2(T_1 - T_2) = T_2 - 20$

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (C,D)**

Given configurative can be though off



Let x current is passing through 1st cylinder and y is passing through 2nd cylinder as shown.

Let us calculate current density (J) = $\frac{I}{A}$

$J = \frac{x}{\pi R^2}$ then.

$I = J \times [\pi R^2 - \pi(R/2)^2]$

$= J \times \frac{3\pi R^2}{4} = \frac{x}{\pi R^2} \times \frac{3\pi R^2}{4} = \frac{3x}{4}$

So $x = \frac{4I}{3}$; similarly $y = \frac{I}{3}$

Magnetic field at A : Due to 1st cylinder + due to 2nd cylinder = 0 + due to 2nd cylinder

$\int H \cdot dl = I_{enclosed}$

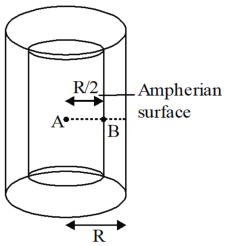
$\Rightarrow H_A 2\pi \frac{R}{2} = \frac{I}{3} \Rightarrow H_A = \frac{I}{3\pi R}$; $B_A = \frac{\mu \cdot I}{3\pi R}$

Magnetic field at B : Due to 1st cylinder + due to 2nd cylinder = Due to 1st cylinder + 0

$\Rightarrow \int H \cdot dl = I_{enclosed} \Rightarrow H_B \int dl = I_{enclosed}$

$H_B \times 2\pi \frac{R}{2} = \left(\frac{4I}{3}\right) \frac{1}{\pi R^2} \times \pi \left(\frac{R}{2}\right)^2 = \frac{I}{3}$

$H_B = \frac{I}{3\pi R} \Rightarrow B_B = \frac{\mu \cdot I}{3\pi R}$



6. **Ans (A,C)**

$\frac{mg\ell}{2} = \frac{1}{2} \cdot I\omega^2$

$\frac{mg\ell}{2} = \frac{1}{2} \cdot \frac{m\ell^2}{3} \cdot \omega^2$

$\frac{3g}{\ell} = \omega^2$

$R_x = m\omega^2 \left(\frac{\ell}{2}\right) = m \cdot \frac{3g}{e} \cdot \frac{\ell}{2} = \frac{3mg}{2}$; $a_y = \frac{\ell}{2}\alpha$

$mg - R_y = ma_y$; $mg\frac{\ell}{2} = \frac{m\ell^2}{3} \cdot \alpha = \frac{3\ell g}{2\ell} = \alpha$

$R_y = mg - m\left(\frac{\ell}{2}\right)\left(\frac{3g}{2\ell}\right)$

$R_y = mg - \frac{3mg}{4} = \frac{mg}{4}$

$R = \sqrt{R_x^2 + R_y^2}$

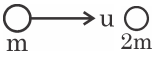
7. **Ans (A,B,C)**

On inserting an iron rod in the solenoid along axis, then the magnetic field associated increase, magnetic flux linked with solenoid increases and self inductance of solenoid increases.

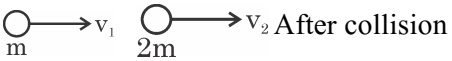
PART-1 : PHYSICS

SECTION-I (iii)

8. **Ans (C)**



Before collision



After collision

$mu = mv_1 + 2mv_2$

$\Rightarrow \frac{V_2 - V_1}{u} = e = 1$

It given $V_1 = -\frac{u}{3}$, $V_2 = \frac{2u}{3}$

So, after collision

$|P_1| = \frac{P}{3}$, $|P_2| = \frac{4P}{3}$

$KE_1 = \frac{K}{9}$ $KE_2 = \frac{8K}{9}$

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11. **Ans (D)**

Apply conservation of angular momentum to get required result

PART-1 : PHYSICS

SECTION-II

1. **Ans (4)**

$-60 \times 2 + 80 \times 1 + 100\ V = 0$

$V = \frac{4}{10} m/s$

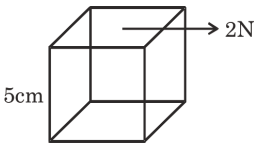
2. **Ans (1)**

Just after closing the switch, no current will flow through inductor.

Current through battery, $I = \frac{100V}{1000\Omega} = 10^{-2}A$

So, $P = VI = 100 \times 10^{-2} = 1W$

3. **Ans (4)**



$\text{Shear strain} = \frac{1 \times 10^{-3}}{5 \times 10^{-2}} = \frac{1}{50}$

$\eta = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{F/A}{\text{Shear strain}}$

$= \frac{2}{25 \times 10^{-4}} \times 50 = 4 \times 10^4 \Rightarrow x = 4$

4. **Ans (5)**

$\sigma eA\,(T^4 - T_0^4) = mS\left(\frac{dT}{dt}\right)$

$\therefore \frac{dT}{dt} = \frac{\sigma eA\,(T^4 - T_0^4)}{mS}$

$= \frac{\sigma e\left(\pi\frac{d^2}{4}\right)(T^4 - T_0^4)}{\rho\frac{4}{3}\pi\left(\frac{d}{2}\right)S}$

On putting values and solving, we get

$\frac{dT}{dt} = 200 \times 10^{-3}\ °C/s$

5. **Ans (3)**

Use $\left(\frac{\mu_0 i}{2R}\right)\frac{\theta}{2\pi}$

6. **Ans (2)**

Purely rolling solid sphere KE $= \frac{7}{10}mv_{cm}^2$

$\frac{7}{10}mv_{cm}^2 = mgh$

$v_{cm} = \sqrt{\frac{10gh}{7}}$

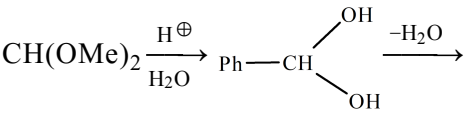
Now $x = \sqrt{\frac{2h}{g}}v_{cm} = \sqrt{\frac{2h}{g}}\sqrt{\frac{10gh}{7}} = \sqrt{\frac{20}{7}}h$

$\frac{x}{h}\sqrt{\frac{7}{5}} = 2$

PART-2 : CHEMISTRY

SECTION-I (i)

3. **Ans (C)**



Ph – CH = O

4. **Ans (B)**

β–keto carboxylic acid liberates CO_2 on heating.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (B,C)**

Reduction and electronation take place at cathode elctrode, so it become positive electrode.

7. **Ans (A,C,D)**

(A) In 0.6 m M HCOOH solution

$K_a = \frac{[H^+]^2}{C - [H^+]} \Rightarrow 8 \times 10^{-4} = \frac{[H^+]^2}{6 \times 10^{-4} - [H^+]}$

$\therefore [H^+] = \mathbf{4 \times 10^{-4}}\ \mathbf{M}.$

So solution in (A), (C) & (D) are ISOHYDRIC.

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PART-2 : CHEMISTRY

SECTION-I (iii)

8. Ans (A)

P) H atom : (P - 4)

$(V_n)_Z = 2.18 \times 10^8 \left(\frac{z}{n}\right)$ cm/ sec

$(V_1)_H = 2.18 \times 10^8$cm/ sec

Q) He^+ $(Q \rightarrow 1)$

$(r_n)_Z = 0.529 \times \frac{n^2}{z} A^0$

$=(r_2)_{He^+} = 0.53 \times \frac{4}{2} A^0 = 0.53 \times 2A^0$

R) $(E_4)_{Be^{3+}} = -13.6 \times \frac{z^2}{n^2}$

$= -13.6 \times \frac{16}{16} = -13.6$eV

S) $(r_3)_{Li^{2+}} = 0.53 \times \frac{9}{3} = 0.53 \times 3A^0$

11. Ans (A)

Stability of alkene $\propto$ number of α - H

Heat of combustion $\propto$ no. of carbon

$\propto \frac{1}{\text{Stability of Alkene}}$

Heat of Hydrogenation $\propto$ No. of π - bond

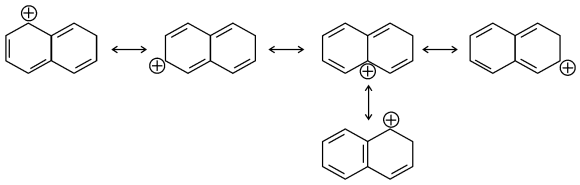
$\propto \frac{1}{\text{Stability of Alkene}}$

Bond order $\propto$ bond energy $\propto \frac{1}{\text{bond length}}$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (5)



2. Ans (9)

$E = -\frac{(I.P.)}{n^2}$

$\Delta E = I.P.\left(1 - \frac{1}{4}\right) = 36 \times \frac{3}{4} = 27$ eV

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$\therefore P^{-1}Q(P^T)^{-1} = A$

or $Q = PAP^T \quad (PT^T = I)$

$\therefore Q^2 = (PAP^T)(PAP^T) = PA^2P^T$

$\therefore Q^{2019} = PA^{2019}P^T$

$\therefore P^TQ^{2019}P = A^{2019}$ or $\begin{bmatrix} 1 & 2019 \\ 0 & 1 \end{bmatrix}$

2. Ans (B)

$x - 3y - 3 = 0$ & $\frac{x^2}{9} - \frac{y^2}{4} = 1$

$(y+1)^2 - \frac{y^2}{4} = 1 \Rightarrow y = 0, -\frac{8}{3}$

$(x, y) = (3, 0) = A$ & $\left(-5, -\frac{8}{3}\right) = B$

$AB = \frac{8\sqrt{10}}{3}$ & $P \equiv \left(-1, \frac{-4}{3}\right)$

3. Ans (C)

$y = \{x\}^{[x]} + [x]$

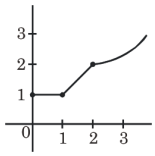
$$y = \begin{cases} 1 & x \in [0,1) \\ x & x \in [1,2) \\ (x-2)^2 + 2 & x \in [2,3) \\ 3 & x = 3 \end{cases}$$

$A = 1 + \frac{3}{2} + \int_2^3 \left((x-2)^2 + 2\right) dx$

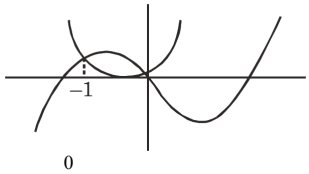
$A = \frac{5}{2} + \left(\frac{(x-2)^3}{3} + 2x\right)_2^3$

$A = \frac{5}{2} + \left(\frac{1}{3} + 6\right) - 4$

$= \frac{29}{6}$



4. Ans (B)



$A = \int_{-1}^{0} (x^3 - 3x - 2x^2) dx$

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PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C,D)

For an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \ (a > b)$

Equation of director circle : $x^2 + y^2 = a^2 + b^2 = R^2$ ....(1)

Equation of auxiliary circle : $x^2 + y^2 = a^2$ ....(2)

Equation of given circle : $4x^2 + 4y^2 = 3a^2$ ....(3)

Let $p(\alpha, \beta)$ on the circle then ....(1)

$\alpha^2 + \beta^2 = a^2 + b^2 = R^2$

$\therefore$ Equation of chord of contact for (2)

$\therefore \ \alpha x + \beta y - a^2 = 0$ ....(4)

If chord of contact touch the circle (3) then by condition of tangency.

$$\left|\frac{-a^2}{\sqrt{\alpha^2+\beta^2}}\right| = \frac{\sqrt{3}a}{2} \text{ or}$$

$$\left(\frac{a^2}{\sqrt{a^2+b^2}}\right)^2 = \left(\frac{\sqrt{3}a}{2}\right)^2$$

$(a^2 + b^2 = \alpha^2 + \beta^2)$

or $4a^2 = 3a^2 + 3b^2$

or $a^2 = 3b^2$

Now for ellipse we know

$\Rightarrow b^2 = a^2(1 - e^2)$

$\Rightarrow e = \sqrt{\frac{2}{3}}$

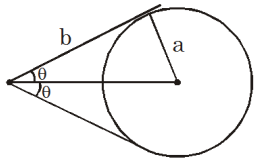
for option (B)

If radius of circle is 8 then $a^2 + b^2 = 64$

or $b = 4 \quad (a^2 = 3b^2)$

or $a = 4\sqrt{3}$

foci $= (\pm ae, \ 0) = (\pm 4\sqrt{2}, \ 0)$



for option 'C' length of tangent $\sqrt{a^2 + b^2 - a^2} = |b|$

for option 'D' $\tan\theta = \frac{a}{b}$

$\tan\theta = \sqrt{3}$

$\theta = 60°$

$\therefore 2\theta = 120°$

6. Ans (A,B,C,D)

Given minor axis is $2y + x = 3$

Equation of major axis $2x - y = k$

Passes through (2, 3) $\quad k = 1$

hence major axis $2x - y = 1$

centre of ellipse (1, 1)

$$\frac{a}{e} = \sqrt{(3-1)^2 + (2-1)^2} = \sqrt{5}$$

$\because \quad b^2 = a^2(1 - e^2)$

$$\therefore \ \frac{5}{4} = 5e^2(1 - e^2) \ \Rightarrow \ e = \frac{1}{\sqrt{2}}$$

$$LR = \frac{\sqrt{5}}{\sqrt{2}}$$

equation of ellipse

$$4(2x - y - 1)^2 + 2(x + 2y - 3)^2 = 25$$

7. Ans (B,D)

$$A^2 \text{adj}(A) = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 2 \\ k & 0 & 2 \end{bmatrix}$$

$$\Rightarrow \ A.A \ \text{adj}(A) = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 2 \\ k & 0 & 2 \end{bmatrix}$$

$$\Rightarrow \ |A|A = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 2 \\ k & 0 & 2 \end{bmatrix}$$

$$\Rightarrow \ |A|^4 = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 2 \\ k & 0 & 2 \end{bmatrix} \Rightarrow \ k = 2$$

$$\Rightarrow A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{bmatrix}$$

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PART-3 : MATHEMATICS

SECTION-I (iii)

8. Ans (A)

(P) If $|A| = 2$, then $\left|2A^{-1}\right| = \frac{2^3}{|A|} = 4$

(Q) $|A| = \frac{1}{8}$ then $|\text{adj (adj}(2A)|$

$\Rightarrow 2^{12}\,|\text{adj (adj}A)| = \frac{2^{12}}{|A|^4} = \frac{2^{12}}{2^{12}} = 1$

(R) If $(A+B)^2 = A^2 + B^2$ then $AB + BA = 0$

or $AB = -BA$ or $|AB| = |-BA|$

or $2|A|\,|B| = 0$ or $|B| = 0$

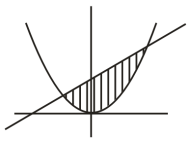
(S) $|A_{2\times 2}| = 2$ $|B_{3\times 3}| = 3$ and $|C_{4\times 4}| = 4$

$\frac{|ABC|}{8} = \frac{|A|\,|B|\,|C|}{8} = \frac{2\times 3\times 4}{8} = 3$

9. Ans (B)

(P) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\sec^2 x - \tan^2 x\right) dx = \frac{\pi}{2}$

(Q) $x = 4y - 2$ and $x^2 = 4y$



$\Rightarrow x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0$

$x = 2, -1$

so, $\int_{-1}^{2} \left(\frac{x+2}{4} - \frac{x^2}{4}\right) dx = \frac{9}{8}$

(R) $2ae = \left|\left(1+\sqrt{2}\right) - \left(1+\sqrt{2}\right)\right| = 2\sqrt{2}$

$ae = \sqrt{2}$

$a = 1$

$\Rightarrow b = 1 \because e = \sqrt{2} \Rightarrow$ Hyperbola is rectangular

$\Rightarrow$ L. R. $= \frac{2b^2}{a} = 2$

(S) $y^2 = 8x + 4y + 4$

$(y-2)^2 = 8(x+1)$

$y^2 = 4ax$

$a = 2, X = x + 1, Y = y - 2$

focus (1,2)

$y - 2 = m(x - 1)$

Put (3,0) in the above line

$m = -1$

Length of focal chord = 16

10. Ans (C)

(P) $\left(1 + \left(\frac{dy}{dx}\right)^2\right)^3 = \lambda\left(\frac{d^2y}{dx^2}\right)^2$,

$\alpha = 2$

$\beta = 2$

$(\alpha + \beta) = 4$

(Q) $y = \tan\left(\frac{\pi}{4} + ax\right)\tan\left(\frac{\pi}{4} - ax\right) + ce^{bx+d}$

$y = \left(\frac{1+\tan ax}{1-\tan ax}\right)\left(\frac{1-\tan ax}{1+\tan ax}\right) + ce^d . e^{bx}$

$y = 1 + A.B^x$, order = 2

(R) $\frac{dy}{dx} = y - y^2 \Rightarrow \frac{dy}{y(1-y)} = dx$

$\log y - \log|1 - y| = x + C$

$\log 2 - \log 1 = 0 + C$

$C = \log_e 2$

$\log y - \log|y - 1| = x + \log_e 2$

$\log\left|\frac{y}{y-1}\right| = \log_e 2 + x$

$\frac{y}{y-1} = 2e^x \Rightarrow y = 2e^x y - 2e^x$

$y = \frac{2e^x}{2e^x - 1} \Rightarrow \lim_{x\to\infty} y = 1$

(S) $x = my + \frac{1}{m}$

$\frac{dx}{dy} = m \Rightarrow m = \frac{1}{y'}$

$x = \frac{y}{y'} + y' \Rightarrow (y')^2 + y = xy'$

$\alpha = 1, \beta = 2$

$\alpha + \beta = 3$

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11. **Ans (A)**

Equation of S_3 is

$(x-0)(x-1)+(y-0)^2+\lambda y=0$

i.e., $x^2+y^2-x+\lambda y=0$

i.e., $\sqrt{\frac{1}{4}+\frac{\lambda^2}{4}-0}=1 \Rightarrow \lambda=\pm\sqrt{3}$

$\therefore$ As circle S_3 lie above x-axis

$\Rightarrow \lambda=-\sqrt{3}$

$\Rightarrow S_3 \equiv x^2+y^2-x-\sqrt{3}y=0$

$\therefore$ centre of $S_3 \equiv \left(\frac{1}{2},\frac{\sqrt{3}}{2}\right)$

slope of line joining C_1 & $C_2=\sqrt{3}$

$\Rightarrow$ Slope of common tangent is $\sqrt{3}$.

i.e., $\sqrt{3}x-y+k=0$, is required

common tangent and touches circle S_1

$\Rightarrow \left|\frac{k}{\sqrt{3+1}}\right|=1 \Rightarrow k=\pm 2$

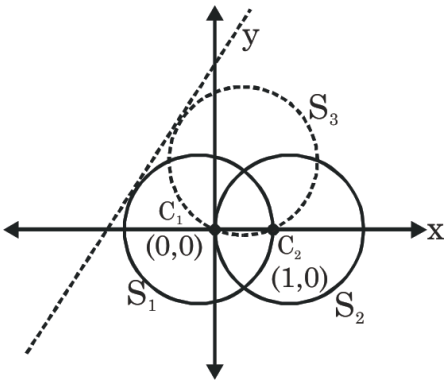
$\therefore k=2$

$\Rightarrow$ Requried common tangent is

$\sqrt{3}x-y+2=0$

$\Rightarrow a=\sqrt{3}, b=-1$

$\therefore a^2=b=3-(-1)=4$



PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1)**

$x\cos(x^2y)\{2xydx+x2dy\}+\sin(x^2y)dx+dy=0$

$\Rightarrow x\cos(x^2y)\,d(x^2y)+\sin(x^2y)dx+dy=0$

$\Rightarrow d(x\sin(x^2y))+dy=0$

$\Rightarrow x\sin(x^2y)+y=c$

as $y(0)=0 \Rightarrow c=0$

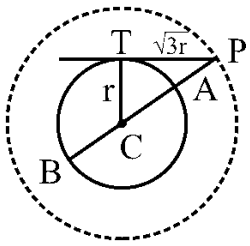
and $y(\alpha)=1 \Rightarrow \alpha\sin\alpha^2=-1 \Rightarrow |\alpha\sin\alpha^2|=1$

2. **Ans (8)**

$\begin{bmatrix}1 & 2\\ 5 & 10\end{bmatrix} \rightarrow$ 4 cases and $\begin{bmatrix}2 & 1\\ 10 & 5\end{bmatrix} \rightarrow$ 4 cases

3. **Ans (8)**

$PA.PB=PT^2$



$\Rightarrow$ $PC=2r$

4. **Ans (7)**

$A=\int_0^1 (2x-2x^2)\,dx-\int_0^1 x\log x\,dx$

$A=\frac{7}{12}$ $\quad\therefore\quad 12A=7$

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5. **Ans (2)**

$\frac{dy}{dx} = \ell n(ex) + e^x$

$\Rightarrow y = x\ell nx + e^x + c$

$\because f(1) = e + 1 \Rightarrow \quad c = 1$

$\Rightarrow f(x) = x\ell nx + e^x + 1$

$\lim\limits_{x\to 0^+} f(x) = 0 + 1 + 1$

6. **Ans (8)**

Given parabola is $x^2 + 4y - 6x + k = 0$

$\Rightarrow (x-3)^2 = -4\left(y + \frac{k-9}{4}\right)$

$\therefore$ equation of directrix is

$y + \frac{k-9}{4} = 1 \Rightarrow y = 1 - \left(\frac{k-9}{4}\right) = -1$

$\Rightarrow k = 17$

$\therefore$ Parabola (1) becomes

$(x - 3)2 = -4(y + 2)$

$\therefore$ vertex = (3,−2) and focus = (3,−3)

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